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Transformation from Knudsen diffusion to facilitated diffusion for CO_2 in confined mass transfer channels of polyimide mixed matrix membranes

Ting Wang [a], Ke Zhang [a], Shi-wen Du [a], Li-guang Wu [a,*], Chun-juan Li [a], Hua-li Chen [a], Hai-Chang Guo [b]

[a] School of Environmental Science and Engineering, Zhejiang Gongshang University, Hangzhou 310018, China
[b] School of Pharmaceutical and Chemical Engineering, Taizhou University, Shifu Avenue 1139, Jiaojiang 318000, China

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ABSTRACT

Incorporation of porous materials in a polymeric membrane is an effective strategy for improving the gas permeability of the resulting mixed matrix membranes (MMMs). However, effective improvement of selectivity of membrane toward CO_2 separation is still a significant challenge due to the insufficient CO_2 capture performance of porous materials. In this study, polyimide (PI)-based MMMs with enhanced CO_2 capture performance were prepared for CO_2 separation. First, an ionic liquid (IL) with a high affinity for CO_2 was immobilized in the inner pore structures of two types of mesoporous silica (SiO_2) molecules to form IL@SiO_2 composites. These composites were used as nanofillers to construct the MMMs. The stable immobilization of IL inside the pores of the mesoporous SiO_2 indicated that the IL@SiO_2 and MMMs both exhibited higher adsorption selectivity for CO_2. The gas solubility coefficient of the MMMs was positively proportional to their maximum gas adsorption capacity, which was determined by the gas adsorption capacity of the PI polymer and nanofillers. More interestingly, the diffusion coefficients of different gases in the MMMs were closely related to the confined mass transfer channels introduced by the nanofillers. The diffusion of CO_2 in the confined mass transfer channels of MMMs was facilitated and N_2 diffusion was slightly inhibited, which increased the diffusion selectivity of the resulting MMMs for CO_2. Combined with Maxwell–Wagner–Sillar model calculations for gas permeation, it was confirmed that IL immobilization transformed the diffusion mode of CO_2 in the confined mass transfer channels of the MMMs from Knudsen diffusion to facilitated diffusion. However, the immobilization of excessive ILs in the mesopores blocked the mass transfer channels of the MMMs, resulting in lower gas permeability and decreased CO_2 separation selectivity.

1. Introduction

The huge consumption of fossil fuels releases a significant quantity of greenhouse gases into the atmosphere, which has led to environmental issues of global concern, such as climate warming and ocean acidification [1–3]. High-performance CO_2 capture and separation technologies, such as low-temperature distillation [4], amine absorption [5], and membrane separation [6], are some of the most effective strategies for reducing carbon emissions. Among them, membrane separation is increasingly used in CO_2 separation processes due to its advantages including convenient fabrication of membrane, low energy consumption, and easy operation [6–8]. The key factor affecting the performance of these membranes is the appropriate selection of membrane materials [9,10]. Owing to their advantages of low operating cost and easy construction of membrane modules, polymers are increasingly utilized to construct CO_2 separation membranes, and exhibit promising practical application prospects [10,11]. However, there exists a well-known "trade-off" between permeability and selectivity in the separation processes of polymer membranes. In other words, if the membrane permeability is high, its selectivity is low, and vice versa [12,13]. Therefore, the permeability and selectivity of most polymer membranes are limited by the Robeson line (2008), which places an upper bound on their performance [14]. At present, a universal and effective strategy for overcoming this "trade-off" effect is the fabrication of mixed matrix membranes (MMMs), which are prepared by incorporating nanomaterials in a polymeric matrix [15–17]. The interaction between the incorporated nanomaterials and CO_2 molecules can enhance the capture of CO_2 by MMMs. Moreover, the interfacial voids formed in MMMs by

* Corresponding author.
E-mail address: wulg64@hotmail.com (L.-g. Wu).

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nanomaterial doping can potentially accelerate the mass transfer of gas molecules. Compared to gas diffusion in polymeric matrices, gas molecules exhibit a faster mass transfer rate in these interfacial voids [15–17]. To date, many zero-dimensional to multidimensional nanomaterials have been employed to fabricate high-performance MMMs for gas separation, such as titania (TiO_2) and silica (SiO_2) nanoparticles [18,19], single-walled and multi-walled carbon nanotubes (CNTs) [20,21], graphene and graphene oxide (GO) [22,23], and microporous or mesoporous molecular sieves (such as zeolites) [24,25]. The addition of nanomaterials with good CO_2 adsorption performance (such as GO and amino-grafted CNTs [21]), can improve the performance of membranes for capturing CO_2 molecules. Moreover, the defects in GO or the tubular structure of CNTs can introduce interfacial voids into polymers, improving the permeability of the resulting MMMs. However, the interfacial voids in MMMs introduced by the addition of conventional inorganic nanomaterial exhibit poor stability and controllability. Voids with sizes exceeding 5 nm or even tens of nm are often formed in these constructed MMMs, resulting in the non-selective permeation of all gas molecules through the MMMs [21,22].

To obtain a stable and structurally controllable interfacial void structure, nanomaterials with pore structures (such as molecular sieves, SiO_2 hollow nanospheres, etc.) have been employed as nanofiller dopants to introduce pore structures with dimensions below 5 nm into the polymeric matrix [26,27]. These small pore structures can act as gas-confined mass transfer channels in MMMs. However, the pore walls of these nanomaterials cannot effectively capture CO_2 molecules, indicating the acceleration of the mass transfer of all gas molecules (including N_2, CH_4, and O_2) in the MMM channels [26,27]. This makes it difficult to facilitate the CO_2 mass transfer process. In our previous study [28], helical mesoporous SiO_2 with a pore size of 3–5 nm was first employed as a nanofiller dopant to introduce an asymmetric gas-confined mass transfer channel in MMMs. The objective of this study was to improve the capture of CO_2 by the pore walls of the mesoporous SiO_2. It was found that the Si–OH groups on the pore surface served as capture sites for CO_2 molecules, which accelerated the mass transfer rate of CO_2 in the channels and thus enhanced the selectivity of the resulting MMMs for CO_2 separation. Moreover, the collision probability between gas molecules and the walls of the helical channels was improved, which hindered the mass transfer of N_2 molecules in these helical channels. However, the capture of CO_2 molecules by this mesoporous SiO_2 via the Si–OH groups on the SiO_2 surface was still insufficient, thus the CO_2 separation selectivity of the MMMs containing helical mesoporous SiO_2 was found to be lower than that of GO-incorporated MMMs, with the same amount of SiO_2 or GO [29,30]. Another study has shown that the CO_2 capture performance of mesoporous SiO_2 can be improved by grafting its inner pore walls with amino groups or via the immobilization of ionic liquid (IL) with good CO_2 adsorption properties [31,32]. In particular, the immobilization of IL in the inner channels of composite SiO_2-based materials is a promising strategy for obtaining excellent CO_2 adsorption performance through simple and convenient methods [31,32]. Therefore, porous materials loaded with IL have been widely used to prepare high-performance CO_2 absorption materials. The Shahram research group found that the adsorption capacity of modified porous materials for CO_2 increased by 4.35 times compared to the unmodified porous materials when imidazole ILs (such as 1-hexyl-3-methylimidazolium bis (trifluoromethylsulfonyl) imide ([hmim][Tf_2N])) were immobilized inside the pores [31,32].

Inspired by previous studies [28,29] and the recent progress in this field [31,32], two widely-used mesoporous SiO_2 (namely, SBA-15 and MCM-41) were selected as carriers to immobilize IL in this study. Noteworthy, SBA-15 and MCM-41 exhibit a large specific surface area and stable ordered mesoporous channels, which make them suitable candidates for immobilizing IL inside the channels. Moreover, SBA-15 and MCM-41 were first selected for IL immobilization, due to their similar characteristics other than pore size, such as surface silicon groups and regular ordered pore structure. In this study, [hmim][Tf_2N] IL was employed for immobilization due to its affinity for CO_2 molecules and good CO_2 absorption capacity [31,32]. After immobilization, polyimide (PI)-based MMMs with high permeation performance for CO_2 separation were fabricated by incorporating IL-immobilized mesoporous SiO_2. Combined with the results in our previous study, a permeation model of the membrane separation of gas molecules was constructed (based on an improved Maxwell–Wagner–Sillar model (MWS)) to describe and predict the different CO_2 and N_2 separation processes in MMMs with incorporated IL-immobilized mesoporous SiO_2. The different effects of IL@SiO_2 on the gas capture performance of the MMMs and the diffusion of gas molecules in the MMMs were distinguished. Furthermore, the influence of the immobilized IL on the diffusion of two different types of gases (CO_2 and N_2) in the mesoporous SiO_2 channels and the confined mass transfer channels in the MMMs was clarified. Finally, the main factors affecting the CO_2 separation performance of the membranes were identified.

2. Experimental

2.1. Materials

3,3′,4,4′-Biphenyltetracarboxylic dianhydride (BPDA), 4,4′-diaminodiphenylether (ODA), and *N*,*N*-dimethylacetamide (DMAC) were purchased from Reagent Chemical Manufacturing (Shanghai, China). Further, [hmim][Tf_2N] (purity 98 %, density 1.379, melting point −9 °C, molecular weight 447.41) was purchased from McLean Reagent Co., ltd.

2.2. Immobilization of IL in two mesoporous silica molecules

The filler was prepared by a facile impregnation method, and a schematic of the preparation process is shown in Fig. S1 (Supporting Information, SI). First, [hmim][Tf_2N] was dissolved in an appropriate amount of acetone in a 25-mL beaker. A certain amount of mesoporous SiO_2 (SBA-15 or MCM-41, as presented in Table S1 in SI) was suspended in this solution, and the mixture was magnetically stirred at room temperature for 24 h. Next, the solvent was separated by filtration. No ILs were found in the filtered clear solution, as revealed by evaporation at 100 °C, indicating that the ILs used in this study could undergo complete immobilization in mesoporous SiO_2. Finally, the obtained solid was dried under vacuum at 60 °C for 12 h to obtain IL-immobilized mesoporous SiO_2 (IL@SiO_2).

2.3. Preparation of MMMs containing mesoporous silica with or without IL immobilization

The fabrication of MMMs with the incorporation of varying amounts of different IL@SiO_2 materials was carried out by *in-situ* polymerization according to a previously described method [33,34]. The thickness of the polymer membranes was about 15 ± 2 μm, which is consistent with that of the MMMs containing GO or MWCNTs reported in our previous studies [28,29]. The relevant details are described in Fig. S2. The added amounts of mesoporous SiO_2 or IL@SiO_2 during the preparation of MMMs were 0.0910 g (1 wt% loading content), 0.1820 g (2 wt% loading content), 0.2785 g (3 wt% loading content), 0.4739 g (5 wt% loading content), and 0.6777 g (7 wt% loading content), respectively.

2.4. Characterization

The morphologies of the prepared materials and membranes were characterized by transmission electron microscopy (TEM) using a Tecnai G2 F20 instrument. Scanning electron microscopy (SEM) images were obtained using a HITACHI SU8010 microscope. The material structures were characterized by Fourier transform infrared (FTIR) spectroscopy (Nexus-670, Nicolet Co.). X-ray photoelectron spectroscopy (XPS) with AlKα X-ray (hm = 1486.6 eV) radiation operated at 150 W (Thermo K

Alpha, USA) was applied to investigate the surface properties of the samples. N_2 adsorption/desorption isotherms were obtained at 77 K using a Quantachrome NOVA 4200E surface area and pore size analyzer. Thermogravimetric analysis (TGA) was performed in a thermal analysis instrument (TG 209 F3, Netzsch, Germany). The mechanical properties of membranes were tested by Tensile tester (INSTRON 5982, Instrang Corporation of America).

2.5. Gas sorption measurements and gas permeability

Independent CO_2 and N_2 sorption measurements were carried out on the powder samples and membranes using a pressure decay sorption system. First, the entire system was degassed overnight. Next, the reservoir cell was filled with a known amount of test gas and thermally equilibrated for 10–15 min. The valve between the reservoir and sample cell was then opened to charge the sample cell with gas, and LabView was used to record the pressure decay in both cells. When the pressure reached equilibrium, the run was stopped and the amount of sorbed gas was calculated by using a mole balance. Pure gas sorption measurements were performed at 25 °C on different membranes.

The pure gas permeability values were determined by the constant-volume/variable-pressure method [35,36]. The gas permeability was determined by using Eq. (1) as follows:

$$P = D \times S = 10^{10} \times \frac{VL}{p_{up}ART} \times \frac{dp(t)}{dt} \quad (1)$$

Where P is the gas permeability (Barrer) [1 Barrer = 10^{-10} cm^3 (STP) cm cm^{-2} $s^{-1}$$cmHg^{-1}$], p_{up} is the upstream pressure (cmHg), dp/dt denotes the change in the steady-state permeate-side pressure (cm•Hg·s^{-1}), V is the calibrated permeate volume (cm^3), L is the membrane thickness (cm), A is the effective membrane area (cm^2), T is the operating temperature (K), and R is the gas constant [0.278 cm^3•cmHg·cm^{-3}(STP) K^{-1}]. The diffusivity (D) was determined by using Eq. (2) as follows:

$$D = L^2/6\theta \quad (2)$$

Where θ is the time lag when a steady dp/dt rate is obtained on the downstream side in the permeation tests [37]. The ideal selectivity (α) was determined by using Eq. (3):

$$\alpha = P_{CO_2}/P_{N_2} \quad (3)$$

3. Results and discussion

3.1. Morphology of different IL@SiO_2 samples

The mesoporous structures of the two silica materials (MCM-41 and SBA-15) with or without IL immobilization were first evaluated by nitrogen adsorption–desorption analysis to determine their Brunauer–Emmett–Teller (BET) surface areas and pore characteristics, as shown in Fig. S3. The unmodified mesoporous silica materials and those with IL immobilization exhibited type IV sorption isotherms with H_2-type hysteresis loops, confirming the presence of mesoporous structures in all materials [38]. The H_2-type hysteresis loops became smaller and their P/P_0 range shifted from 0.40 to 0.8 to 0.2–0.4 after IL immobilization. The specific surface areas, pore sizes, and pore volumes of both silica materials before and after IL immobilization are listed in Table 1. Notably, IL immobilization led to a slight decline in the specific surface area, pore size, and pore volume of both types of mesoporous SiO_2 materials. This result indicates that the immobilized IL entered the pores and was attached to the inner pore walls of the mesoporous SiO_2.

Table 1
Textural characteristics of both mesoporous SiO_2 with or without IL immobilization, as determined by nitrogen adsorption–desorption analysis (Added contents of IL and mesoporous SiO_2 are 0.25 g and 1.0 g, respectively).

Sample	Surface area/m^2 g^{-1}	Pore volume/cm^3 g^{-1}	Pore size/nm
MCM-41	1028.5	0.89	3.5
IL@MCM-41	886.6	0.71	3.2
SBA-15	508.1	1.23	9.7
IL@SBA-15	323.8	0.79	9.4

TEM images of both mesoporous SiO_2 materials with or without IL immobilization were obtained, as shown in Figs. S4 and S5. Both mesoporous SiO_2 materials exhibited a clear and regularly ordered pore structure. After IL immobilization, the overall morphology of the mesoporous SiO_2 did not change significantly. However, the boundaries of the internal pore structure of both mesoporous SiO_2 materials became slightly blurred after IL immobilization, which was attributed to the intrusion of IL into the pores.

The FTIR spectra of both mesoporous silica materials with or without IL immobilization are shown in Fig. S6. A wide peak (~3500 cm^{-1}) corresponding to the tensile vibration of the O—H bond on the surface of the mesoporous silica was observed for both silica materials. This peak became weaker after IL immobilization, which indicates that the IL was immobilized by reacting with the hydroxyl groups on the silica surface. New peaks are also observed in the FTIR spectra of both silica samples after IL immobilization. Peaks at 2882, 2975, and 3188 cm^{-1} are attributed to the symmetric and asymmetric stretching vibrations of aromatic C—H groups on the $[hmim]^+$ cations of IL. The peaks at 1583, 1472, and 1355 cm^{-1} correspond to the stretching vibrations of the C═N, C═C, and C—N bonds in the $[hmim]^+$ cation of IL, respectively [39]. Moreover, peaks at 1208, 1141, and 1063 cm^{-1} are attributed to the stretching vibrations of the C–F, S—N—S, and S═O bonds in the $[Tf_2N]^-$ anion of IL. These peaks indicate the successful immobilization of IL [hmim][Tf_2N] on both MCM-41 and SBA-15, respectively. The TGA curves displayed in Fig. S7 show that both mesoporous SiO_2 materials without IL immobilization did not experience significant mass loss in the temperature range from room temperature to 800 °C. In contrast, both IL@MCM-41 and IL@SBA-15 exhibited significant mass loss at 500 °C. About 20 % of the sample mass was lost in both cases, which was similar to the amount of IL added during the preparation process. These results indicate that IL was completely immobilized in both MCM-41 and SBA-15. The temperature range of mass loss in the TGA curve of IL@MCM-41 (100–500 °C) was significantly wider than that in the TGA curve of IL@SBA-15 (100–400 °C). This was potentially because MCM-41 consisted of smaller pores and a larger specific surface area than SBA-15, leading to stronger protection of the IL inside the silica channels of MCM-41.

The XPS profiles of both mesoporous silica materials with or without IL immobilization are shown in Fig. S8. Three clear peaks at 690.1, 401.2, and 234.2 eV correspond to the F, N, and S elements in IL, respectively [40]. The N 1s spectra of IL@MCM-41 and IL@SBA-15 (Fig. S9) show two peaks at binding energies of 399.9 and 402.3 eV, which are attributed to the imide in the $[Tf_2N]^-$ ion and the C—N—C in imidazole ring of the $[hmim]^+$ cation of the IL, respectively. These peaks confirm that the IL [hmim][Tf_2N] was successfully introduced in both mesoporous SiO_2 materials. The O 1s spectra of both mesoporous silica materials with or without IL immobilization are shown in Fig. 1. The two significant peaks at 525.2 and 532.9 eV are attributed to Si–O–Si and Si–OH, respectively. After IL immobilization, the peak at 532.9 eV became significantly small or even disappeared, which indicates that the hydrogen bonds formed by the interaction between [hmim][Tf_2N] and mesoporous SiO_2 led to the decrease in the concentration of Si–OH on the surface of the mesoporous SiO_2. This is in good agreement with the FTIR spectroscopy results.

3.2. Gas adsorption by IL@SiO_2 samples

The adsorption of CO_2 and N_2 by the prepared silica samples with or without IL immobilization was evaluated, as shown in Fig. 2. The maximum amount of gas adsorbed by each sample (Q_m) was calculated through gas adsorption curves by using Langmuir model (detailed

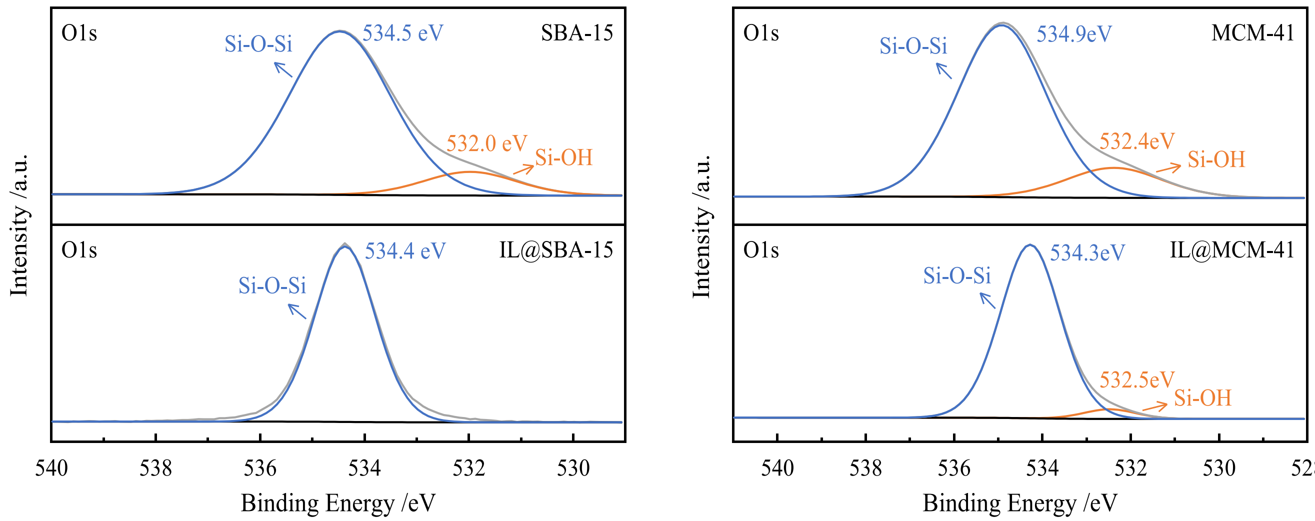

Fig. 1. High-resolution O 1s XPS profiles of both silica samples with or without IL immobilization (added contents of IL and mesoporous SiO_2 are 0.25 g and 1.0 g, respectively).

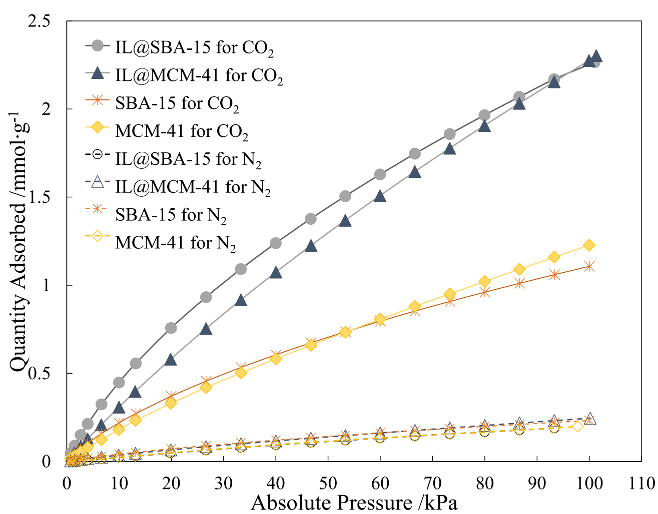

Fig. 2. Gas adsorption curves and corresponding Langmuir adsorption isotherms of both mesoporous SiO_2 with or without IL immobilization (added content of IL and mesoporous SiO_2 are 0.25 g and 1.0 g, respectively).

description is given in Supporting Information), which has been well accepted to describe the gas adsorption by IL or other adsorbents. These results presented in Table 2 demonstrate that CO_2 and N_2 adsorption conformed to Langmuir adsorption isotherms for all materials. Both silica materials exhibited much better CO_2 adsorption capacity than N_2 adsorption capacity. This was attributed to the better interaction between the CO_2 molecules and the silica hydroxyl groups on the surface of mesoporous silica. IL immobilization decreased the hydroxyl groups on the surface of the mesoporous silica. Nevertheless, the excellent CO_2 adsorption capacity of the IL improved the adsorption performance of both IL-immobilized mesoporous silica materials for CO_2, with IL immobilization almost doubling the CO_2 adsorption capacities of the unmodified MCM-41 and SBA-15. However, interaction was not observed between the IL and N_2 molecules; as a result, the N_2 adsorption performance of the mesoporous SiO_2 materials did not significantly change after IL immobilization. This result indicates that IL immobilization could effectively enhance the CO_2 adsorption capacity of the mesoporous silica. Thus, it was speculated that this could enhance the CO_2 capture performance of the resulting MMM containing IL-immobilized mesoporous silica.

Table 2
Maximum adsorption capacity Q_m of mesoporous SiO_2 samples obtained from the Langmuir adsorption isotherms.

Sample	Q_m/mmol g^{-1}		R^2	
	CO_2	N_2	CO_2	N_2
IL@SBA-15	4.71	0.86	0.99826	0.99967
IL@MCM-41	8.92	0.83	0.99988	0.99912
SBA-15	2.20	0.53	0.99817	0.99855
MCM-41	4.14	0.86	0.9991	0.99966

3.3. Morphology of PI MMMs incorporating IL@SiO_2 materials

Figs. 3 and S10 show SEM images of the surface of MMMs prepared with both mesoporous SiO_2 materials with or without IL immobilization. For comparative analysis, SEM images of neat PI and IL-incorporated MMMs were also obtained, as displayed in Fig. 3. No serious agglomeration was observed in the MMMs prepared with any of the silica samples, indicating well dispersion of both mesoporous SiO_2 materials in the polymeric matrix even after IL immobilization. However, when IL was directly added to the PI polymer, serious agglomeration was observed, as shown in Fig. 3f. This result indicates that the IL immobilized in the pore channels of the mesoporous SiO_2 was well-protected by the SiO_2. Noteworthy, after *in-situ* polymerization to fabricate the MMMs, the IL was still stable and immobile in the pore channels of the mesoporous SiO_2. Therefore, the two IL-immobilized mesoporous SiO_2 materials maintained good dispersion in the MMMs. For further systematic characterization of the morphology of MMMs, high-magnification SEM images of IL@MCM-41/PI and IL@SBA-15/PI (with 5.0 wt% nanofiller content) were also obtained, as presented in Fig. S11. In addition, SEM images and high-magnification SEM images of cross-section of IL@MCM-41/PI and IL@SBA-15/PI (with 5.0 wt% nanofiller content) were also listed in Fig. S11. From these SEM images, it is revealed that most nanofillers could be uniformly dispersed in the MMMs (with 5.0 wt% nanofiller content), although a few aggregations were also observed in SEM images (as shown by high-magnification SEM images of the surface and cross-section of IL@SBA-15/PI).

Furthermore, the morphology of nanomaterials in MMMs was characterized by TEM, and the TEM images of MMMs containing MCM-41 and SBA-15 (membrane slicing) are shown in Fig. S12. The results indicate that both MCM-41 and SBA-15 exhibited good dispersion in polymers, but still the orientation of mesoporous SiO_2 in the MMMs

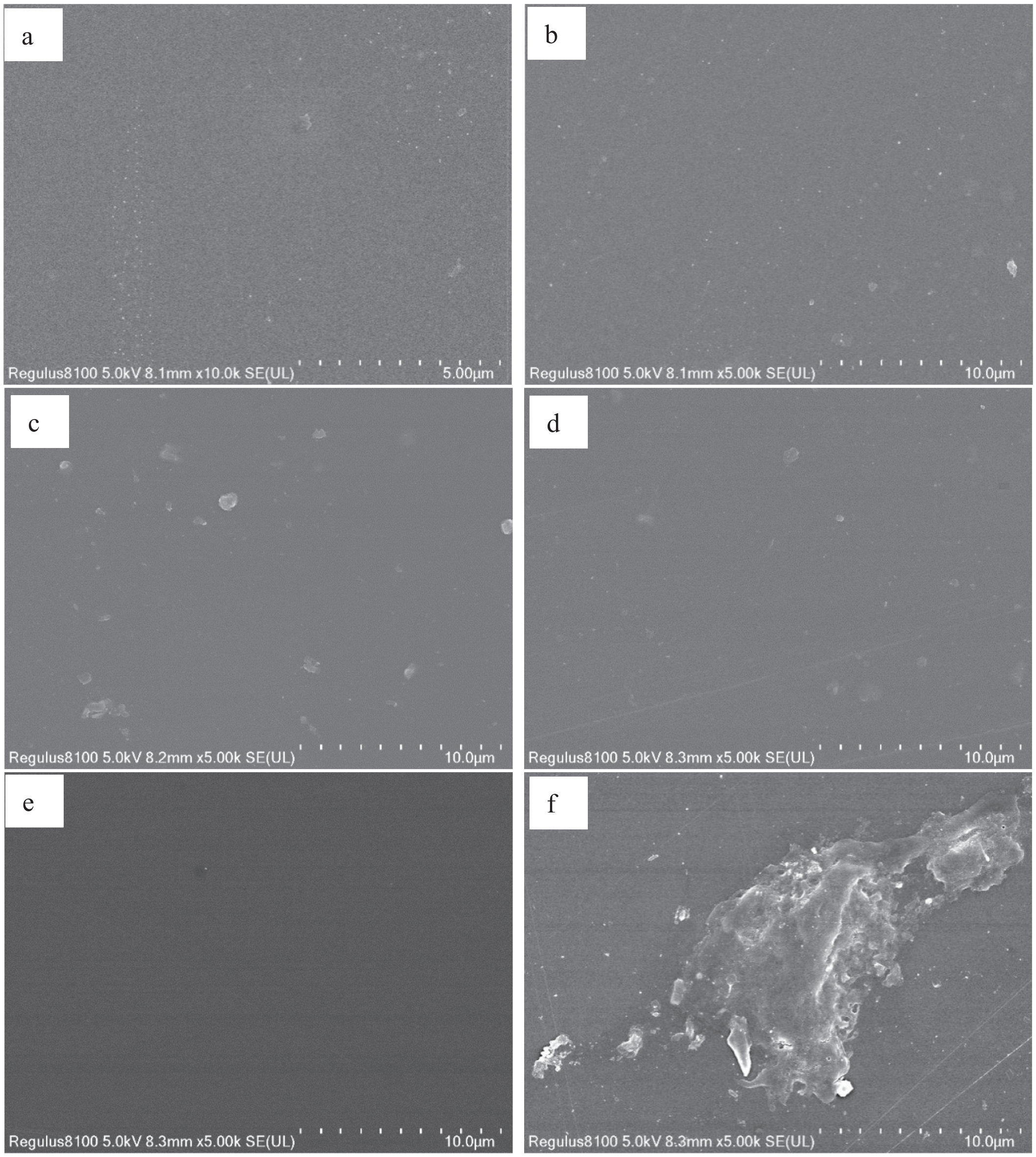

Fig. 3. SEM images of PI and different MMMs: a. IL@MCM-41/PI (with 3.0 wt% nanofiller content); b. IL@SBA-15/PI (with 3.0 wt% nanofiller content); c. IL@MCM-41/PI (with 5.0 wt% nanofiller content); d. IL@SBA-15/PI (with 5.0 wt% nanofiller content); e. neat PI; and f. IL/PI (with 5.0 wt% of IL).

could not be determined. TEM images of MMMs show that the addition of mesoporous SiO_2 could introduce a regular and ordered pore structure into the polymer.

3.4. Separation performance of PI MMMs prepared with IL@ SiO_2.

The mechanical strength of PI, MCM-41/PI, and IL@MCM-41/PI was measured, and the results are presented in Fig. S13 and Table S2. It was found that the mechanical strength of MCM-41/PI and IL@MCM-41/PI was superior to that of PI, which indicates that the addition of MCM-41 and IL@MCM-41 could enhance the mechanical properties of MMMs.

3.4.1. CO_2 and N_2 gas adsorption curves and isotherms of PI and MMMs

The CO_2 and N_2 gas adsorption curves and corresponding Langmuir isotherms of PI and different MMMs are shown in Fig. S14, and their maximum adsorption capacities are listed in Table 3. The adsorption process of these samples also conformed to Langmuir adsorption isotherms. The addition of either type of IL-immobilized mesoporous silica material could obviously improve the CO_2 adsorption performance of the MMMs, but did not significantly affect their N_2 adsorption performance. Compared to neat PI, the adsorption performance of MMMs for CO_2 separation increased by about 10.6 % after adding 5.0 wt% of IL@SBA-15, while addition of 5.0 wt% IL@MCM-41 led to the increase in the adsorption performance of MMMs toward CO_2 separation by about 26.2 %. Furthermore, the theoretical maximum CO_2 and N_2 adsorption capacities of the MMMs Q'_m were calculated by using the IL@SiO_2 adsorption data presented in Table 2 and the doping amount of IL@SiO_2 (1 and 5 wt%). These calculated capacities are also listed in Table 3. The Q_m values of the MMMs with IL@SiO_2 (the doping amount of IL@SiO_2 was 1 and 5 wt%, respectively) for both CO_2 and N_2 adsorption were close to the theoretical Q'_m values, which further confirms that both IL@SiO_2 materials were uniformly distributed in the PI polymeric matrix. Moreover, the gas adsorption performance of the PI polymers and IL@SiO_2 determined the gas capture ability of the MMMs.

3.4.2. Permeation performance of PI and MMMs for CO_2 and N_2

The CO_2 and N_2 permeabilities of MMMs prepared with varying amounts of each mesoporous SiO_2 material with or without IL

Table 3
Maximum adsorption capacity Q_m of PI and different MMMs obtained from their Langmuir adsorption isotherms.

Sample	Experimental Q_m/mmol•g^{-1}	*Calculated Q'_m/mmol•g^{-1}
PI for CO_2	2.63	/
PI for N_2	0.86	/
IL@MCM-41 (1.0 wt%)/PI for CO_2	2.75	2.69
IL@MCM-41 (1.0 wt%)/PI for N_2	0.82	0.86
IL@MCM-41 (5.0 wt%)/PI for CO_2	3.31	3.57
IL@MCM-41 (5.0 wt%)/PI for N_2	0.92	0.89
IL@SBA-15 (1.0 wt%)/PI for CO_2	2.71	2.65
IL@SBA-15 (1.0 wt%)/PI for N_2	0.88	0.86
IL@SBA-15 (5.0 wt%)/PI for CO_2	2.91	3.05
IL@SBA-15 (5.0 wt%)/PI for N_2	0.91	0.89

$^*Q'_m = (Mass\,fraction\,of\,PI) \times Q_{m,\ PI} + (Mass\,fraction\,of\,IL@SiO_2) \times Q_{m,\ IL@SiO_2}$

immobilization (0–7 wt%) are shown in Fig. S15, and their CO_2/N_2 ideal selectivities are displayed in Fig. 4. Addition of the mesoporous SiO_2 materials significantly improved the CO_2 permeability of the resulting MMMs, which was due to the good CO_2 adsorption ability of SiO_2 and IL@SiO_2. In particular, the MMMs containing IL@MCM-41 exhibited higher CO_2 permeability than the MMMs containing IL-free SiO_2 and IL@SBA-15 with the same filler content. The N_2 permeability of the MMMs was significantly affected by IL immobilization. Incorporation of MCM-41 and SBA-15 in the PI polymer led to significant improvement in the N_2 permeability of the MMMs, with permeability even exceeding the CO_2 permeability of MCM-41/PI and SBA-15/PI. However, the N_2 permeability of the MMMs prepared with IL@MCM-41 and IL@SBA-15 exhibited less of an improvement than the CO_2 permeability. The differences in the CO_2 and N_2 separation performance of the MMMs incorporating mesoporous SiO_2 with or without IL immobilization were further confirmed by evaluating their CO_2/N_2 ideal selectivities (Fig. 4). Incorporation of MCM-41 and SBA-15 in PI lowered the CO_2/N_2 ideal selectivities of the resulting MMMs. With the increase in the doping amount of both mesoporous silica materials, the decline in the CO_2/N_2 ideal selectivity became more pronounced. Interestingly, the CO_2 adsorption capacities of MCM-41 and SBA-15 were larger than their N_2 adsorption capacities, as shown in Fig. 2. Therefore, the decline in the CO_2 separation selectivity of MCM-41/PI and SBA-15/PI was caused by the weaker CO_2 diffusion selectivity of the MMMs after the addition of MCM-41 and SBA-15, as shown in Fig. S16 and Fig. 5. This was attributed to the Knudsen diffusion of CO_2 and N_2 molecules in the MMMs containing MCM-41 and SBA-15, which caused the lighter N_2 molecules to diffuse faster than the CO_2 molecules.

It is well known that the diffusion of gas molecules in pores follows the Knudsen diffusion mechanism when the mean free path of the gas molecules is much larger than the pore size of the diffusion channel. In this study, the mean free paths of CO_2 and N_2 gas molecules were 62.9 and 60.0 nm, respectively (in the standard state) [41]. These mean free paths were much larger than the pore sizes of MCM-41 (3.5 nm) and SBA-15 (9.6 nm). Therefore, the diffusion of both gas molecules in the mesoporous channels of the MMMs followed the Knudsen mechanism, and the diffusion selectivity of CO_2/N_2 in the mesoporous channels was only 0.8, according to the Knudsen diffusion equation (see SI for the calculation process). After the incorporation of MCM-41 or SBA-15 into the PI matrix of the membranes, the CO_2 separation selectivity of the resulting MMMs decreased. Moreover, incorporation of an increasing amount of either mesoporous SiO_2 material led to a more pronounced decline in selectivity. However, Figs. 4 and 5 illustrate that the MMMs containing IL@MCM-41 and IL@SBA-15 exhibited significantly stronger ideal selectivity and diffusion selectivity for CO_2 separation than neat PI and the MMMs prepared using mesoporous silica without IL immobilization. When less than 7 wt% SiO_2 was used, the selectivity of the MMMs containing IL@MCM-41 or IL@SBA-15 for CO_2 separation increased with increasing mesoporous silica content.

The Knudsen diffusion process of gas molecules in mesoporous channels occurs through collisions between the gas molecules and the walls of the channels, as shown in Fig. S17. After IL immobilization, the IL on the channel walls is unfavorable for collision of gas molecules with the walls, as shown in Fig. S17. Therefore, the diffusion performance of

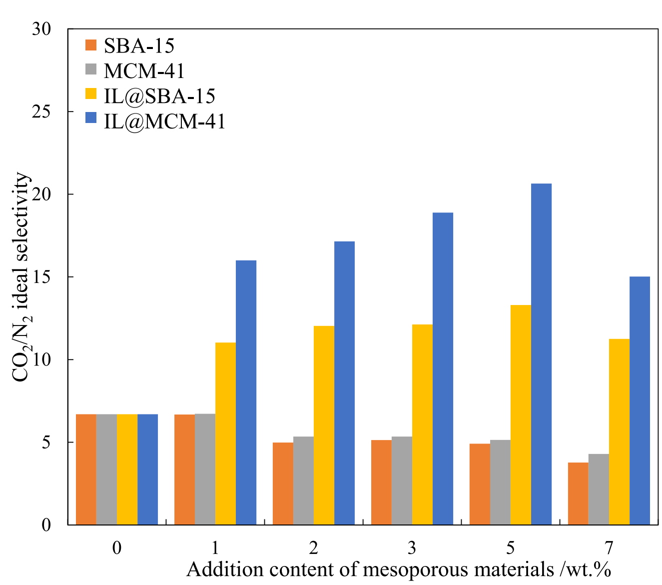

Fig. 4. Ideal CO_2/N_2 selectivity of PI and different MMMs.

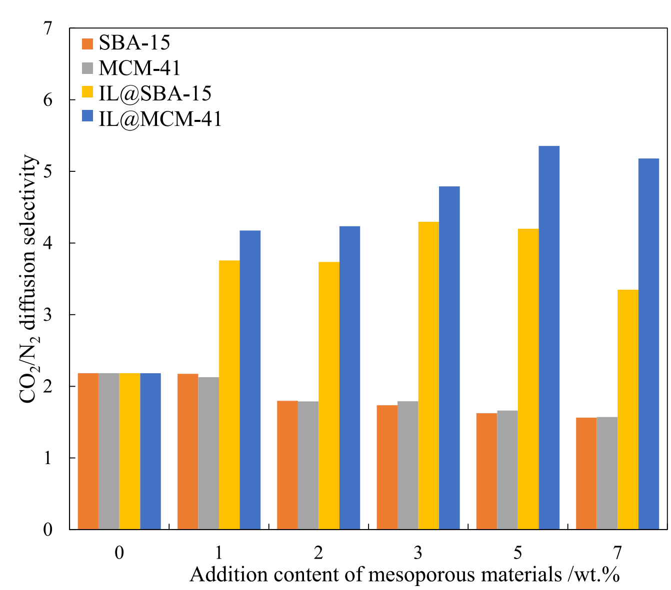

Fig. 5. CO_2/N_2 diffusion selectivity of PI and different MMMs.

N_2 molecules through IL@SiO_2/PI MMMs was weaker than that through SiO_2/PI MMMs, under the same conditions. In contrast, the IL immobilized on the inner pore walls promoted the mass transfer of CO_2 molecules in the pores due to the interaction between the IL and CO_2 molecules, and the diffusion mechanism of CO_2 got transferred from Knudsen diffusion to facilitated diffusion. Therefore, the Knudsen diffusion of both the CO_2 and N_2 molecules in the pores was greatly weakened. The CO_2 diffusion process was improved, while the N_2 diffusion process was suppressed, which significantly enhanced the CO_2/N_2 diffusion selectivity of the MMMs. Moreover, addition of IL@MCM-41 and IL@SBA-15 increased the CO_2 capture performance of the MMMs. Therefore, the CO_2/N_2 ideal selectivity of the MMMs was significantly enhanced by the addition of well-distributed IL@MCM-41 or IL@SBA-15. However, when more than 7 wt% of IL@MCM-41 or IL@SBA-15 was added, significant agglomerates were found on the surface of the MMMs (Fig. S18), which negatively affected the CO_2 separation performance.

To confirm the effect of the diffusion process of CO_2 and N_2 molecules after IL immobilization, CO_2 and N_2 gas penetration experiments using MCM-41 and IL@MCM-41 were conducted to evaluate the gas diffusion process, as shown in Fig. S19. The figure illustrates that the time for CO_2 gas to reach the maximum gas concentration is significantly shortened after IL immobilization, which can verify that IL immobilization accelerates the speed of CO_2 gas molecules passing through mesoporous materials. In contrast, because IL on the pore wall hindered the collision between N_2 molecules and the walls, the penetration of N_2 molecules is hindered after IL immobilization, which means that the speed of N_2 molecules passing through the powder slowed down. This result confirms that the IL immobilization favored the facilitated diffusion for CO_2 in the mesoporous channels of IL@MCM-41.

The separation performance of different MMMs with the upper bound line (2008) is presented in Fig. S20. Clearly, the separation performance of MMMs containing 5 wt% IL@MCM-41 was close to the upper bound (2008). Therefore, IL immobilization is a promising strategy to enhance the overall separation performance of MMMs. IL immobilization with a small amount can obviously enhance the performance of MMMs for CO_2 separation.

3.5. Mechanism of enhanced performance for CO_2 separation and gas permeation model

The effect of adding mesoporous SiO_2 with or without IL immobilization on the CO_2 separation performance of the MMMs was further elucidated by evaluating the CO_2/N_2 ideal selectivity α, diffusion selectivity α_D, solubility selectivity α_S, and adsorption selectivity α_Q ($\alpha_Q = Q_{m,CO_2}/Q_{m,N_2}$, where Q_{m,CO_2} and Q_{m,N_2} are the maximum adsorption capacities of neat PI and MMMs for CO_2 and N_2, respectively), which are listed in Table 4. The addition of SBA-15 or MCM-41 improved the selectivity toward CO_2 solubility in the as-fabricated MMMs. Owing to the Knudsen diffusion of CO_2 and N_2 molecules in the mesoporous channels of both SBA-15 and MCM-41, the CO_2 diffusion selectivities of SBA-15/PI and MCM-41/PI were significantly lower than that of the neat PI. Therefore, the addition of SBA-15 or MCM-41 reduced the ideal selectivity of MMMs toward CO_2. After adding the IL-immobilized mesoporous silica materials, the resulting MMMs showed a significant increase in both CO_2 solubility selectivity α_S and diffusion selectivity α_D. Interestingly, the selective adsorption selectivity α_Q and solubility selectivity α_S values of the IL@SBA-15/PI and IL@MCM-41/PI MMMs were comparable, which indicates that $\alpha_Q = Q_{m,CO_2}/Q_{m,N_2} = S_{CO_2}/S_{N_2} = \alpha_S$. This result reveals that the solubility coefficients of the gas molecules in the MMMs are positively correlated with their adsorption performance.

To further explain the differences in the CO_2 diffusion process in the MMMs prepared with mesoporous SiO_2 materials with or without IL immobilization, a mathematical gas permeation model was constructed. The doping of MCM-41 and IL@MCM-41 was used to describe the permeation or transport of gas molecules through the MMMs. According to our previous study [28], MCM-41 and IL@MCM-41 in a polymeric matrix should be regarded as collections of parallel channels instead of simple spheres or cylinders, as shown in Fig. S21. The MWS model (a general and valid model for predicting the permeation of gas through MMMs loaded with non-spherical fillers) was employed to further predict the gas transport through the MMMs containing MCM-41 and IL@MCM-41 [42–44]. In the MWS model, the formula for general permeability is described as follows:

$$\frac{P_{MMMs}}{P_{PI}} = [\frac{nP_s + (1-n)P_{PI} - (1-n)\varphi(P_{PI} - P_s)}{nP_s + (1-n)P_{PI} + n\varphi(P_{PI} - P_s)}] \tag{4}$$

Where P_{MMMs} is the effective permeability of an MMM, P_{PI} and P_S are the permeabilities of the continuous phase (the PI polymer) and dispersed phase (the nanofiller), respectively, and φ denotes the volume fraction of the dispersed phase in the MMM. The parameter n precisely takes the shape and connectivity of the domains into account. Considering the relationship between gas permeability coefficient and solubility and diffusion coefficients (P=S × D), Eq. (5) can be written as follows:

$$\frac{S_{MMMs} \bullet D_{MMMs}}{S_{PI} \bullet D_{PI}} = [\frac{n \bullet (S_s \bullet D_s) + (1-n)\bullet(S_{PI} \bullet D_{PI}) - (1-n) \bullet \varphi \bullet (S_{PI} \bullet D_{PI} - S_s \bullet D_s)}{n(S_s \bullet D_s) + (1-n)(S_{PI} \bullet D_{PI}) + n \bullet \varphi \bullet (S_{PI} \bullet D_{PI} - S_s \bullet D_s)}] \tag{5}$$

Where S_{MMMs} is the gas solubility coefficient of an MMM, and S_{PI} and S_S are the gas solubility coefficients of the continuous and dispersed phases, respectively. D_{MMMs} represents the gas diffusion coefficient of the MMM, and D_{PI} and D_S denote the gas diffusion coefficients of the continuous and dispersed phases, respectively. Currently, the gas solubility coefficient S_S of the dispersed phase (mesoporous silica) is difficult to measure. The data presented in Table 3 show that the gas solubility coefficient of each membrane is positively proportional to the maximum adsorption capacity Q_m. Therefore, it is assumed that:

$$S_S = \frac{Q_{m,s}}{Q_{m,PI}} \bullet S_{PI} \tag{6}$$

Table 4
CO_2/N_2 selectivities of PI and MMMs.

Selectivity for CO_2/N_2	PI	SBA-15/PI(5 wt%)	IL@SBA-15/PI(1 wt%)	IL@SBA-15/PI(5 wt%)	MCM-41/PI(5 wt%)	IL@MCM-41/PI(1 wt%)	IL@MCM-41/PI(5 wt%)
α	6.70	4.91	11.03	13.30	5.15	15.44	20.60
α_D	2.19	1.62	3.76	4.20	1.66	4.17	5.35
α_S	3.06	3.03	2.93	3.17	3.10	3.68	3.85
α_Q	3.06	/	3.07	3.20	/	3.48	3.92

Where $Q_{m,s}$ and $Q_{m,PI}$ are the Q_m values for gases in the nanofillers and PI polymer, respectively. Therefore, the relationship among S_{MMMs}, S_{PI}, and S_S is:

$$S_{MMMs} = (1-\omega_s)\bullet S_{PI} + \omega_s \bullet S_s = [1+\omega_s \bullet (\frac{Q_{m,s}}{Q_{m,PI}} - 1)]\bullet S_{PI} \tag{7}$$

Where ω_s is the silica weight fraction. After substituting Eq. (6) into Eq. (5):

$$\frac{D_{MMMs}}{D_{PI}}\bullet\left[1+\omega_s\bullet\left(\frac{Q_{m,s}}{Q_{m,PI}}-1\right)\right]\bullet S_{PI} = [\frac{n\bullet\left(\frac{Q_{m,s}}{Q_{m,PI}}\bullet D_s\right)+(1-n)\bullet(D_{PI})-(1-n)\bullet\varphi\bullet\left(D_{PI}-\frac{Q_{m,s}}{Q_{m,PI}}\bullet D_s\right)}{n\bullet\left(\frac{Q_{m,s}}{Q_{m,PI}}\bullet D_s\right)+(1-n)(\bullet D_{PI})+n\bullet\varphi\bullet\left(D_{PI}-\frac{Q_{m,s}}{Q_{m,PI}}\bullet D_s\right)}] \bullet S_{PI} \tag{8}$$

In this study, the volume fraction φ of MCM-41 was estimated by using the pure component densities [45] as follows:

$$\varphi = \frac{\omega_s}{\omega_s + \frac{\rho_s}{\rho_{PI}}(1-\omega_s)} \tag{9}$$

Where ρ_s and ρ_{PI} denote the densities of silica and pure PI polymer, respectively, and ω_s is the nanofiller weight fraction.

The parameters in Eq. (8) are listed in Table S3 for an MMM prepared with MCM-41 as the nanofiller. Details of how these parameters were obtained are also reported in the SI. For CO_2 separation, Eq. (8) can be written as:

$$D_{MMMs} = \frac{1.824 + 4.668\omega_s}{(1.52 - 0.32\omega_s)\bullet(1+0.57\omega_s)} \tag{10}$$

For N_2 separation, Eq. (8) can be written as:

$$D_{MMMs} = \frac{0.391 + 2.690\omega_s}{(0.71 - 0.14\omega_s)\bullet(1+0.30\omega_s)} \tag{11}$$

The effect of the amount of MCM-41 doped in the MCM-41/PI membrane on the CO_2 and N_2 diffusion coefficients was determined by using Eqs. (10) and (11), and the results are shown in Fig. 6. To provide a clearer comparative analysis, the experimentally determined diffusion coefficients were also obtained, as shown in Fig. 6.

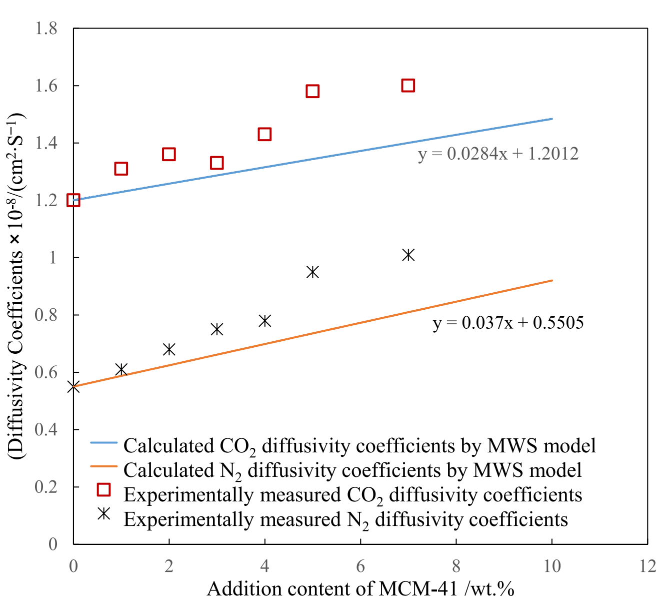

Fig. 6. The effect of MCM-41 doping content on the calculated and experimental CO_2 and N_2 diffusion coefficients of MCM-41/PI.

Fig. 6 shows that the calculated results obtained by using the MWS model for low doping content (less than 5 wt%) are in good agreement with the experimental values, further confirming the reliability of our hypothesis and the suitability of the MWS model in describing the gas permeation process of the prepared MMMs. However, when the loading content of MCM-41 exceeded 5 wt%, in particular, to 7 wt%, the experimental and theoretical values significantly deviated. This was potentially due to the aggregation of MCM-41 in the polymeric matrix.

To further investigate the influence of IL immobilization on the CO_2 mass transfer process in the MMMs, MCM-41 samples with different amounts of immobilized IL were used as nanofillers to fabricate MMMs with 5 wt% nanofiller content (Table S4). The gas adsorption and permeation values of these MMMs (including the maximum adsorption capacity Q_m and CO_2 diffusion coefficient D) were also measured. These values are also listed in Table S4. The addition of the MCM-41 nanofillers with varying content of immobilized IL led to a significant enhancement in the CO_2 capture performance and CO_2 diffusion in the MMMs. The IL was immobilized inside the MCM-41 pores; therefore, the volume fractions φ of the IL@MCM-41 nanofillers were the same as those of unfilled MCM-41. The gas diffusion coefficients D_S of the different nanofillers were calculated by using Eq. (8), as presented in Table 5. The immobilization of 0.1 g (IL immobilization at low content) or 0.25 g (IL immobilization at high content) of IL facilitated the transport of CO_2 molecules inside the MCM-41 pore structure, as shown in Fig. 7. However, after immobilizing 0.5 g of IL (excess IL addition), most of the MCM-41 pores were filled with IL, which negatively impacted the transport of CO_2 molecules. Moreover, the CO_2 diffusion coefficient of this nanofiller was close to the CO_2 diffusion coefficient in the IL phase.

4. Conclusions

An IL with a high affinity for CO_2 was first immobilized in the inner pore structures of two mesoporous silica materials (MCM-41 and SBA-15) to form IL@SiO_2 composites (IL@MCM-41 and IL@SBA-15, respectively). These nanocomposites were used as nanofillers to construct PI mixed matrix membranes (MMMs) for CO_2 separation. Owing to the interaction between CO_2 molecules and the silanol groups of silica, the MMMs prepared with the incorporation of MCM-41 and SBA-15 exhibited good CO_2 adsorption selectivity. However, the mean free paths of CO_2 and N_2 gas molecules were much larger than the pore sizes of both mesoporous silica materials. It indicates that the diffusion of both types of gas molecules in the mesoporous channels of the MMMs followed the Knudsen mechanism, indicating that lighter gas molecules (N_2) diffused more rapidly than heavier gas molecules (CO_2). Therefore, the CO_2 diffusion selectivity of the MMMs decreased after the incorporation of MCM-41 or SBA-15. When IL was immobilized inside the pores of the silica materials, the good affinity between the IL and CO_2 molecules improved the CO_2 adsorption selectivity of both IL@SiO_2 and the

Table 5
The effect of IL immobilization content on Q_m and gas diffusion coefficient of MMMs.

Sample	IL content/ g	$D_{s,CO_2}/\times 10^{-8}$ $cm^2\bullet S^{-1}$	$D_{s,N_2}/\times 10^{-8}$ $cm^2\bullet S^{-1}$
MCM-41	0	3.45	4.30
IL@MCM-41-0.10	0.10	38.55	3.62
IL@MCM-41-0.25	0.25	49.42	3.81
IL@MCM-41-0.40	0.40	42.94	4.16
IL@MCM-41-0.50	0.50	37.53	4.63

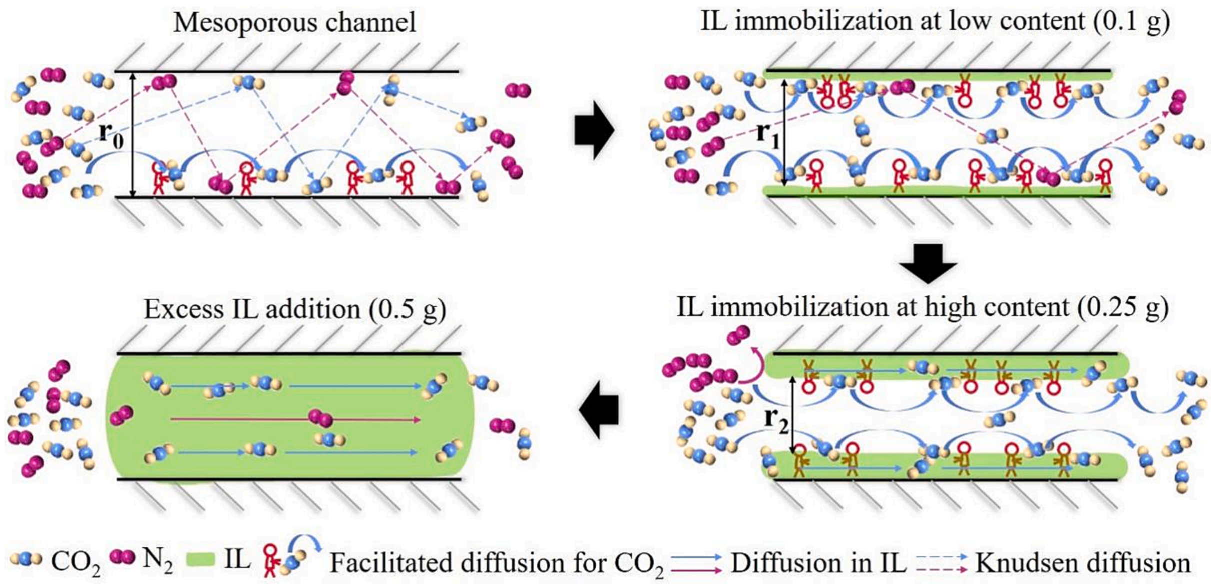

Fig. 7. Transformation of CO_2 transport from Knudsen diffusion to facilitated diffusion in confined mass transfer channels.

resulting MMMs. More interestingly, IL immobilization also enhanced the CO_2 diffusion selectivity of the resulting MMMs, indicating the transformation from Knudsen diffusion to the facilitated diffusion of CO_2.

After dividing the gas permeation through the membranes into solution and diffusion processes, the gas solubility coefficients of the MMMs were found to be positively proportional to their maximum gas adsorption capacity Q_m (calculated from their Langmuir adsorption isotherms). The value of Q_m was determined by using the gas adsorption capacities of the PI polymer and nanofiller in each MMM. Combined with the MWS model calculations for gas permeation, it was found that the diffusion coefficients of CO_2 and N_2 in the MMMs were closely related to the confined mass transfer channels introduced by the nanofillers. The immobilization of IL in the pore structure of MCM-41 or SBA-15 facilitated the diffusion of CO_2 in the confined mass transfer channels of the MMMs, but slightly inhibited N_2 diffusion. Thus, the diffusion selectivity of the resulting MMMs for CO_2 was enhanced. The MWS model calculation results also confirmed that after IL immobilization, the CO_2 diffusion mode in the confined mass transfer channels of the MMMs was transformed from Knudsen diffusion to facilitated diffusion.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2023.147280.

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